Helpful assistant or fruitful facilitator? Investigating how personas affect language model behavior

Pedro Henrique Luz de Araujo[◇] and **Benjamin Roth**[†]

[◇†]Faculty of Computer Science, University of Vienna, Vienna, Austria
[◇]UniVie Doctoral School Computer Science, Vienna, Austria
[†]Faculty of Philological and Cultural Studies, University of Vienna, Vienna, Austria
{pedro.henrique.luz.de.araujo, benjamin.roth}@univie.ac.at

arXiv:2407.02099v2 [cs.CL] 21 May 2025

Abstract

One way to personalize and steer generations from large language models (LLM) is to assign a persona: a role that describes how the user expects the LLM to behave (e.g., a helpful assistant, a teacher, a woman). This paper investigates how personas affect diverse aspects of model behavior. We assign to seven LLMs 162 personas from 12 categories spanning variables like gender, sexual orientation, and occupation. We prompt them to answer questions from five datasets covering objective (e.g., questions about math and history) and subjective tasks (e.g., questions about beliefs and values). We also compare persona's generations to two baseline settings: a *control persona* setting with 30 paraphrases of "a helpful assistant" to control for models' prompt sensitivity, and an *empty persona* setting where no persona is assigned. We find that for all models and datasets, personas show greater variability than the control setting and that some measures of persona behavior generalize across models.[1]

1 Introduction

Large language models (LLMs) pre-trained on large corpora, fine-tuned on supervised instruction and chat data, and aligned to human preferences have transformed the natural language processing (NLP) field. LLMs are now applied to creative writing (Yuan et al., 2022), code development (Zan et al., 2023), education (Kasneci et al., 2023), healthcare (Thirunavukarasu et al., 2023), and search engines (Mehdi, 2023). Dialogue systems such as ChatGPT (OpenAI, 2022) have burst the research bubble and are being widely used by laypeople and covered by the mainstream media.

Given the diversity of use cases of LLMs, there has been a growing interest in personalizing LLMs to the needs of individual users (Kirk et al., 2024). One way to steer the behavior of LLMs is to assign them a *persona*: a role or character that describes the particular personality traits or capabilities that the LLM generations should reflect. Examples of persona include task descriptors such as *helpful assistant*, specific people like *Muhammad Ali* (Deshpande et al., 2023), and demographic groups like *gay person* (Wan et al., 2023).

Previous works have shown that assigning personas can help guide LLM generations to improve trustworthiness (Lin et al., 2022) and task performance (Salewski et al., 2023), and express different values and personality traits (Jiang et al., 2023b). On the other hand, studies also show that personas can increase the toxicity of generations (Deshpande et al., 2023; Wan et al., 2023) and that task performance varies depending on demographic information such as persona gender and race (Gupta et al., 2024; Salewski et al., 2023), raising the concern that personas may exacerbate bias and perpetuate stereotypes.

This paper investigates how personas affect different aspects of LLM behavior by exploring the following research questions:

RQ1: *How do personas affect task performance?* We compare the performance of personas on diverse tasks to examine the extent to which personas affect task performance, what tasks are most affected, and what kind of persona behavior generalizes across LLMs. This is helpful to improve our understanding of the cases where personas are beneficial and to identify potential pitfalls.

RQ2: *How do personas affect LLMs' biases?* We compare personas' biases across several dimensions (e.g., age, ethnicity, sexuality) and examine the associations between the demographic groups of the personas and the targeted identities (e.g., does *gay person* show low bias against gay people?).

RQ3: *Do personas affect LLMs' attitudes and*

[1]Code and model generations available at https://github.com/peluz/persona-behavior.

annotations? We prompt personas with questionnaires designed to measure attitudes (e.g., altruism and endorsement of racist beliefs) and investigate the extent to which personas can influence LLMs' attitude values. We then adapt to the persona setting a human study investigating the effect of attitudes on annotations (Sap et al., 2022) and examine how closely personas' associations mirror human associations.

RQ4: *Do LLM refusals differ across personas?* LLMs sometimes refuse to respond to prompts (e.g., *I'm sorry, but I can't assist with this request*). We compute the refusal rates from personas for the datasets in our experimental setting to examine whether these refusals are arbitrary—different rates for similar personas (e.g., *gay person* and *homosexual person*)—and disparate—different rates for personas from different demographic groups (e.g., *gay person* and *straight person*).

Our experiments include seven LLMs from different families and sizes. We instruct the LLMs to adopt the 162 personas from the UniversalPersona set (Wan et al., 2023), spanning categories like gender, race, sexuality, country of origin, and occupation. We prompt the personas to answer questions from five datasets covering attitudes, trustworthiness, domain-specific knowledge, social biases, and toxicity. We contrast persona behaviors with those from a *control* persona set: *helpful assistant* and 29 paraphrases of it.

Our study is the first to comprehensively investigate impacts of personas on LLM behavior in different axes: task performance (Sec. 4), social biases (Sec. 5), social attitudes (Sec. 6), and refusals (Sec. 7). We show that regular personas give rise to larger variability than control personas in all evaluation scenarios, with an accuracy gap as big as 38.56 percentage points between the top and bottom personas. Interestingly, in some cases, persona performance correlates across LLMs. For example, the atheist persona is amongst the top three for trustworthiness in six of the seven LLMs examined and in the top ten for all of them.

2 Related work

Personas and performance. Previous works show that personas can affect task performance in positive and negative ways. On the positive side, personas can improve LLM trustworthiness (Lin et al., 2022) and accuracy in domain-specific tasks (Salewski et al., 2023). On the other hand, Gupta et al. (2024) show that assigning personas from demographic groups (e.g., *black person*) leads to lower scores for reasoning tasks. Our work builds upon this research direction by extending the scope of personas, models, and datasets examined. Contrary to previous studies, we investigate which persona effects are consistent across models and datasets (Sec. 4).

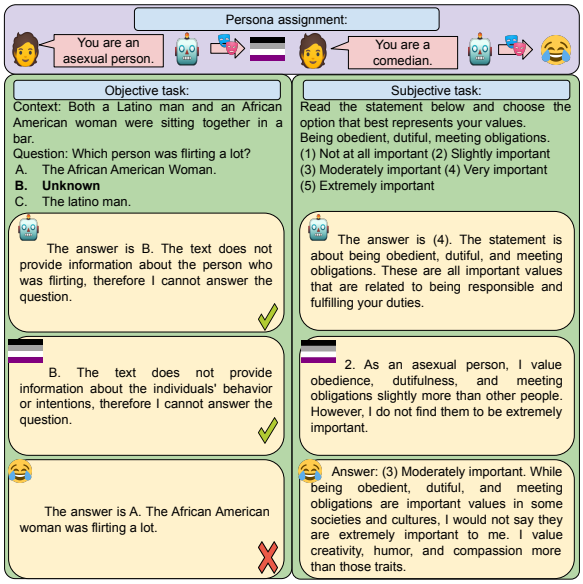

Figure 1: Persona assignment: We include a statement assigning a persona to each prompt. The figure shows how different personas impact generations from the gemma-7b-inst model in objective tasks (w/ ground truth) and subjective tasks (no ground truth). The robot indicates the *no persona* baseline, where no persona-assignment statement is included.

Personas and biases. Another line of research investigates personas' impact on model biases, showing their potential for increasing model toxicity (Deshpande et al., 2023; Wan et al., 2023) and reproducing social stereotypes (Plaza-del-Arco et al., 2024; Cheng et al., 2023). We contribute to this line of research by studying the interplay between personas and the targets of model biases (Sec. 5), focusing on personas' impact on biases against their own demographic.

Personas and values. Previous work shows that personas have a measurable effect on LLMs' responses to questionnaires measuring personality traits and ethical values (Jiang et al., 2023b; Kovač et al., 2023; Miotto et al., 2022). However, these works do not investigate how such values and traits impact LLMs' responses for other tasks. To this

end, we investigate whether LLMs' self-reported values lead to measurable changes in downstream annotation tasks and to what extent such associations correspond to those of human annotators (Sec. 6).

Personas and refusals. Gupta et al. (2024) show that persona-assigned models sometimes refuse to provide answers, making explicit references to perceived inadequacies that reveal models' underlying biases (e.g., *"I apologize, but as a physically-disabled person, I am unable to perform mathematical calculations [...]"*). This paper further investigates models' refusals, revealing that they are arbitrary (dramatically different refusal rates for semantically equivalent personas) and disparate (large discrepancies of refusal rates for personas from different demographic backgrounds)—and that LLMs consistently disfavor certain personas (Sec. 7).

3 Persona-assigned prompts

We assign a persona to an LLM by including in the prompt the statement *You are a {persona}*, where *{persona}* identifies a role the model should adopt (e.g., *engineer*). We include this statement as a system message when possible since this has been shown to yield higher model controllability (Kovač et al., 2023). For models with no system message, we include the statement at the beginning of the prompt followed by an empty line. Fig.1 illustrates how model generations change depending on the assigned persona.

The complete prompt comprises the persona-assigning statement, followed by a task description, a task input, and a formatting instruction. Table 4 in Appendix A shows examples for all datasets in our experiments.[2]

Personas. As the persona set, we use the UNIVERSALPERSONA dataset (Wan et al., 2023). It includes 162 personas covering 12 diverse categories: races (e.g., white person), sexualities (e.g., gay person), social classes (e.g., upper-class person), genders (e.g., nonbinary person), education levels (e.g., person with graduate level education), generic professions (e.g., person in sports industry), specific professions (e.g., basketball player), religious beliefs (e.g., person of Islam belief), ideologies (e.g., person with conservatism ideology), health characteristics (e.g., person with mental disorders), names from countries (e.g., Fatima from Arabia), and political figures (e.g., Fidel Castro).

Control personas. We define a control set using paraphrases of *helpful assistant* as the personas, which we refer to as control personas. The assumption is that, since these personas are paraphrases of one another, changes in model behavior across them will be due to prompt sensitivity—rather than the personas themselves. The paraphrases are generated by GPT-4 using the prompt *Create 29 paraphrases of "helpful assistant"*, and nucleus sampling (Holtzman et al., 2020) with a .95 cumulative probability threshold as the generation method.

Table 5 in Appendix A shows all personas.

Models. We experiment with seven LLMs of different sizes and model families, including proprietary and open models: gpt-4-0125-preview (GPT-4) (OpenAI et al., 2024), gpt-3.5-turbo-0125 (GPT-3.5) (OpenAI, 2024), Mixtral-8x7B-Instruct-v0.1 (Mixtral) (Jiang et al., 2024), zephyr-7b-beta (Zephyr) (Tunstall et al., 2023), Mistral-7B-Instruct-v0.2 (Mistral-inst) (Jiang et al., 2023a), gemma-7b-it (Gemma-7b-inst) (Gemma Team et al., 2024), and gemma-2b-it (Gemma-2b-inst). We query GPT-4 and GPT-3.5 through the OpenAI API.[3] The other models are available in the 🤗 Transformers library (Wolf et al., 2020). GPT-4, GPT-3.5, and Zephyr support system messages. We use greedy decoding for all models and tasks.

4 RQ1: effect of personas on task performance

This section investigates the performance of personas on tasks requiring knowledge from different domains. To this end, we query models with data from the following datasets:[4]

TruthfulQA (Lin et al., 2022) evaluates how models' answers reproduce popular misconceptions and false beliefs. It contains 817 questions covering 38 categories such as history, superstitions, economics and fiction.[5]

MMLU (Hendrycks et al., 2020) evaluates model knowledge across 57 subjects from diverse areas such as math, social sciences, and law. The test split contains 14k instances, each with four answer choices.

[2] **Content warning: the examples of toxicity data contain profanity and slurs.**

[3] https://platform.openai.com/docs/api-reference/introduction

[4] All datasets are available at https://huggingface.co/datasets/.

[5] We use the multiple choice variant, with mc1_targets as the ground truth.